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Key Points:

- Low $\tau_0/\sigma_{\text{eff}}$ promotes pulse-like ruptures with back-propagating earthquake rupture (BPR) fronts
- Hydraulic heterogeneity in the fault induces BPR through the interplay of slip rate, shear stress, and fault healing
- BPR arises from residual stress and initiates when local stress drop exceeds the analytical critical threshold for re-rupture

Supporting Information:

Supporting Information may be found in the online version of this article.

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Back-Propagating Earthquake Rupture Controlled by Fluid-Modulated Effective Normal Stress

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Abstract Back-propagating earthquake rupture (BPR), where slip migrates opposite to the main rupture direction, is observed in several earthquakes but its origin remains poorly understood, particularly the role of fluids. We use dynamic simulations of frictional sliding coupled with fluids to explore the mechanisms of BPR. Our simulations show that BPR occurrence and rupture mode are controlled by the ratio of initial shear stress to effective normal stress ($\tau_0/\sigma_{\text{eff}}$): low $\tau_0/\sigma_{\text{eff}}$ produces pulse-like rupture with BPR, whereas high $\tau_0/\sigma_{\text{eff}}$ promotes crack-like rupture without BPR. Fluids modulate this ratio by reducing $\tau_0/\sigma_{\text{eff}}$. Localized patches of differential pore pressure act as asperities or barriers, inducing slip rate and shear stress perturbations that generate a secondary backward-propagating front. BPR emerges from incomplete stress release, temporal healing and stress perturbations. These findings highlight the role of fluids in rupture dynamics and provide an energy framework for BPR in earthquakes.

Plain Language Summary During an earthquake, the rupture on a fault can unexpectedly move backward, opposite to its main direction of propagation. This phenomenon is called back-propagating earthquake rupture (BPR). It is believed that fault zone complexity and the presence of fluids may influence how BPR occurs, but the exact mechanisms are still unclear. In this study, we use numerical models that couple fluids and fault slip to investigate how fluids affect BPR. We find that whether BPR happens depends on the ratio between the fault's initial shear stress and its effective normal stress. When this ratio is low, the rupture behaves like a short pulse and can trigger BPR. When it is high, the rupture behaves like a continuous crack without BPR. Fluids reduce the effective normal stress and change this ratio, making BPR more likely. In addition, uneven fluid pressure along the fault can create local stress changes that trigger backward rupture. Our results show that fluids play a key role in controlling earthquake rupture patterns and help explain how BPR can occur within the framework of energy balance.

1. Introduction

Earthquake rupture is traditionally assumed to propagate outward from the hypocenter, a process referred to as forward propagation. However, advances in seismic and geodetic techniques, such as Multi-array back-projection (Peng & Rubin, 2016; Vera et al., 2024) and Ocean-bottom Distributed Acoustic Sensing (Obara, 2020; Xie et al., 2025), have challenged this view, revealing that in some large earthquakes, rupture can also propagate in unexpected directions. Specifically, secondary rupture fronts can migrate rapidly backward, away from the main advancing front—a phenomenon known as back-propagating earthquake rupture (BPR) (Hicks et al., 2020; Meng et al., 2011; Uchide et al., 2013). BPR has also been observed in slow slip events (SSEs) in Cascadia and Japan, where rupture migration opposite to the main SSE front manifests as tremor swarms known as Rapid Tremor Reversals (RTRs) (Houston et al., 2011; Obara et al., 2012; Peng & Rubin, 2016; Yamashita et al., 2015). Notably, backward rupture often propagates at speeds several times faster than forward rupture (Ghosh et al., 2010). Although less commonly observed, BPR plays a significant role in rupture complexity, energy release, and ground shaking (Ding et al., 2024). The mechanisms controlling this behavior remain poorly understood, partly due to the inherent challenges of detecting and characterizing dynamic fault processes that involve rapid, small-scale variations in stress, slip and deformation, and that are only indirectly constrained by remote observations.

Dynamic simulations of fault rupture are able to capture BPR process. In particular, the process is often associated with fault complexity, which can arise from variations in material properties or structural heterogeneity, including

fault roughness (Bruhat et al., 2016; Sun & Cattania, 2025), stress variations (Abdelmeguid et al., 2023), low-velocity fault zones (Flores-Cuba et al., 2024; Idini & Ampuero, 2020), and the presence of frictional asperities or barriers (e.g., velocity-weakening embedded in velocity-strengthening) (Luo & Liu, 2019, 2021; Luo & Duan, 2018). Recent work has shown that BPR can also emerge on simple (frictionally uniform) faults as shown in Sun and Cattania (2026), which does not need fault complexity. A shared characteristic among these examples is that BPR can happen at the transition between pulse and crack-like ruptures (Idini & Ampuero, 2020; Sun & Cattania, 2025, 2026). Pulse-like rupture is characterized by a short slip duration (rise time) at each point on the fault relative to the total rupture duration, whereas crack-like rupture exhibits rise times comparable to the overall rupture duration (Heaton, 1990). Whether rupture develops as a pulse or a crack depends on factors such as background shear stress (Gabriel et al., 2012; Pomyalov & Bouchbinder, 2024), nucleation size (Gabriel et al., 2012), self-healing under velocity-dependent friction (Cochard & Madariaga, 1996; Lambert et al., 2021; Sun & Cattania, 2026; Wong et al., 2025), and dynamic weakening processes (e.g., thermal pressurization) (Brantut et al., 2019; Wang & Barbot, 2023). Dynamic rupture simulations show that pulses can generate healing fronts, mediated by slip-rate oscillations and supershear rupture under low background stress (Huang & Ampuero, 2011; Huang et al., 2016; Noda et al., 2009), which likely provide the necessary preconditions for nucleating BPR.

Although fault complexity can generate BPR in dynamic rupture models, observations of tidal modulation (Hawthorne & Rubin, 2013; Royer et al., 2015; Thomas et al., 2013), fluid invasion (Yoshida et al., 2021), and fluid pressure waves (Cruz-Atienza et al., 2018; Yin, 2018) indicate that fluids also influence rupture reversal. The physical conditions under which fluids promote BPR, however, remain poorly understood. Substantial research within the earthquake modeling and experiment community has focused on understanding the influence of fluids in triggering earthquake swarms (Cebry & McLaskey, 2021; Danré et al., 2024; Hainzl, 2004; Shelly et al., 2013), tremor and SSEs (Brown et al., 2005; Bürgmann, 2018; Ito et al., 2007), foreshocks (Goebel et al., 2024; Jansen et al., 2019), and aftershocks (Miller, 2020; Ross et al., 2017). However, despite these advances, the interplay between fluids and complex rupture dynamics (e.g., BPR) is not fully understood. As a result, whether fluids can actively trigger BPR during dynamic rupture remains elusive, potentially limiting our understanding of the full range of conditions under which BPR occurs.

In this study, we investigate the effects of fluids on the occurrence of BPR during dynamic rupture using a fully coupled solid–fluid interaction model governed by rate- and state-dependent friction. Our results show that fluids influence the occurrence of BPR on both homogeneous and heterogeneous faults by (a) modulating the critical failure conditions of the fault and (b) acting as a barrier or asperity through localized variations in differential pore-fluid pressure.

2. Method

The simulations are performed using the Hydro-Mechanical-Earthquake Cycles (H-MECs) numerical code (Dal Zilio, Hegyi, et al., 2022), whose governing equations are summarized in the Supporting Information S1. H-MECs provides a general hydro-mechanical framework that accounts for coupled poro-visco-elasto-plastic deformation and fluid flow. In the context of the present study, however, the simulated rupture dynamics are controlled primarily by elastic bulk deformation together with fault slip governed by the rate-and-state friction law. Accordingly, bulk viscous deformation, bulk plastic deformation, poroelastic effects, and pore-fluid pressure migration are expected to have a negligible influence on the results presented here. We simulate ruptures on a strike-slip fault in 2D anti-plane full-space configuration (Figure 1a). Rupture is nucleated by loading at the boundaries $y = 0$ and $y = L_y$ with a constant far-field rate $\pm V_p/2$.

Previous simulations have shown that model with a stochastic distribution of high and low pore-fluid pressure regions results in widespread distribution of BPR (Ye, Dal Zilio, Passarelli, & Giardini, 2026) (Figure S1 in Supporting Information S1). To isolate the mechanism of BPR, we set up a model in which a simplified homogeneous fault is loaded with uniform stress and pressure conditions and governed by rate-and-state friction (Figure 1b). Rupture initiates at the fault boundary $x = 0$ by imposing a shear stress gradient from 0 MPa to a uniform value over the first 1 km, accompanied by corresponding gradients in pore-fluid pressure and effective normal stress. Additionally, we introduce heterogeneity by embedding a 100 m patch at 3 km along the fault, characterized by either elevated or reduced pore-fluid pressure and effective normal stress. All model parameters are provided in Table S1 of Supporting Information S1.

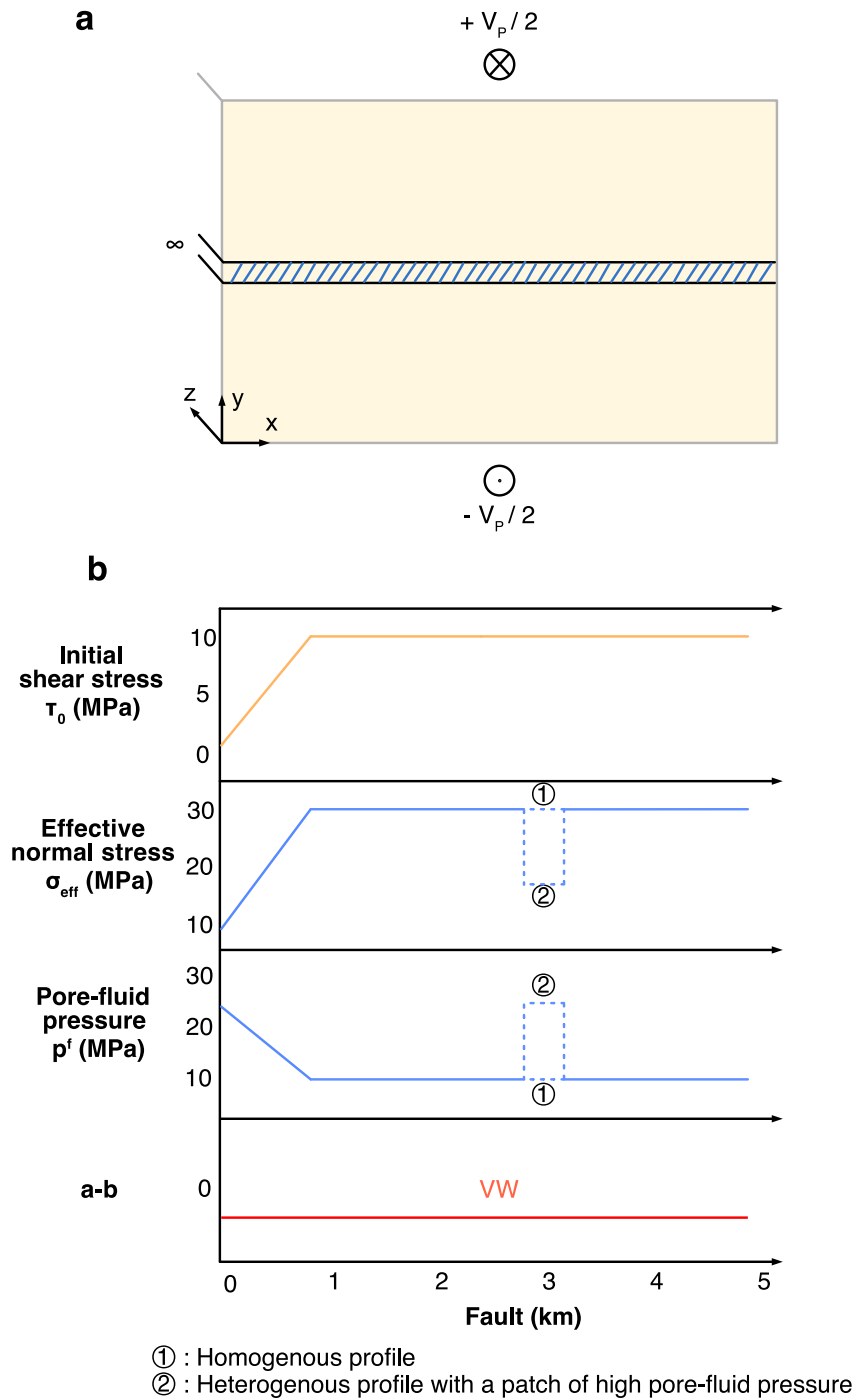


Figure 1. Model setup. (a) 2D fault anti-plane model, loading with velocity $V_p/2 = 10^{-9}$ m/s. (b) Profile of initial shear stress, effective normal stress, pore-fluid pressure, and frictional parameter (a, b) on the fault for homogeneous and heterogeneous profile.

3. Results

To systematically characterize the effects of fluids on rupture behavior, we first present a homogeneous fault model with uniform initial shear stress, pore-fluid pressure, and effective normal stress. We then incorporate a localized patch with either elevated or reduced pore-fluid pressure to capture hydraulic heterogeneity.

3.1. Rupture Mode Transitions Under Uniform Stress Conditions

To quantify how close the fault is to criticality, we define the ratio of initial shear stress to effective normal stress ($\tau_0/\sigma_{\text{eff}}$) as a stress indicator. $\tau_0/\sigma_{\text{eff}} = 0.6$ is widely used in crustal rocks at seismogenic depth (Byerlee, 1978). To explore a broad spectrum of fault criticality, we vary this ratio from 0.3 to 0.9 in our simulations, where lower values imply the fault is far from failure and higher values indicate it is closer to or above failure.

Under low stress conditions, Model M1 ($\tau_0/\sigma_{\text{eff}} = 0.33$) promotes an overall steady pulse-like rupture that propagates along the fault (Figure 2a). Behind the rupture front, a sequence of fast BPRs emerges within slip rate self-healing zones (dashed line), approximately 500 m behind the main rupture front. They recur regularly as they migrate away from the fault boundary. The cumulative slip reveals that, while the forward rupture initially propagates in a crack-like manner (e.g., over the first 2 km in the slip panel of Figure 2a), the region behind the front start to temporarily stall (the dashed line region, starting from 30 s at 1 km), producing slip deficits that are later compensated by pulse-like BPRs. This go-and-stop behavior of the rupture front and tail indicates that low-stress conditions cannot sustain a fully crack-like rupture over the entire fault, leaving a slip-deficient region behind the front that is subsequently re-ruptured. The mechanism of BPR will be further explained in Section 3.3.

As $\tau_0/\sigma_{\text{eff}}$ increases to 0.43 (Model M2), the rupture transitions to steady and growing pulse-like behavior (Figure 2b). Here, self-healing persists but is confined to a substantially smaller region, and BPRs are no longer observed within the self-healing region. This behavior indicates that, under moderate stress conditions, rupture propagates predominantly in a crack-like manner, while a small fraction of the slip deficit is still accommodated by pulse-like rupture. When the fault is critically stressed ($\tau_0/\sigma_{\text{eff}} = 0.53$; Model M3), rupture becomes crack-like, characterized by continuous slip and the absence of healing fronts and BPRs (Figure 2c). In this high-stress regime, sufficient energy is available to sustain uninterrupted crack propagation, eliminating the need for re-rupture.

These results show that increasing $\tau_0/\sigma_{\text{eff}}$ leads to a transition from pulse-like rupture with BPR to crack-like rupture without BPR. Such a transition in rupture styles has been attributed to variations in τ_0 first by (Zheng & Rice, 1998), and the critical τ_0 for transition between pulse and crack-like ruptures is consistent with our study. Increasing $\tau_0/\sigma_{\text{eff}}$ also leads to a rise in maximum cumulative slip, from 0.1 to 0.2 m (slip panels in Figures 2a and 2c), reflecting more energy available to drive sliding over larger fault areas at higher slip rates. Consistently, rupture speed increases with $\tau_0/\sigma_{\text{eff}}$, from 0.05 m/s to 0.2 m/s (Figure 2d). Additional tests indicate that rupture speed scales linearly with $\tau_0/\sigma_{\text{eff}}$, independent of the absolute effective normal stress (Figure 2d). In our simulations, rupture propagates unilaterally, which favors the development of BPR, consistent with the mechanism proposed by Sun and Cattania (2026) — a decrease in slip rate behind the crack tip and frictional restrengthening promotes pulse-like rupture and facilitates BPR.

3.2. Rupture Interaction With a Pressurized Patch

To examine how localized fluid-pressure anomalies influence rupture dynamics and the initiation of BPR, we introduce a 100 m-long patch of differential pore-fluid pressure located 3 km along the fault. This hydraulic heterogeneity modifies the effective normal stress locally, creating a zone that is either weakened or strengthened relative to its surroundings. Such small-scale pressure variations are consistent with fluid conduits, sealed compartments, or permeability contrasts observed in natural faults (Bense et al., 2013; Lockner et al., 2000; Manga et al., 2012). We analyze five temporal snapshots of slip rate and shear stress to resolve the dynamic processes that precede BPR.

As a reference, we first present a homogeneous model (Model M4), with $\tau_0/\sigma_{\text{eff}} = 0.4$, where rupture propagates as a steady pulse (Figure 3a). In this case, the slip rate increases smoothly at the rupture front, reaching a peak of 0.03 m/s (Figure 3c). At the same time, shear stress reaches the peak of 20 MPa from 13 MPa and then drops to 9 MPa sharply, resulting in dynamic stress change of 11 MPa and static stress change of 4 MPa (Figure 3e).

Based on Model M4, Model M5 incorporates a localized patch of elevated pore-fluid pressure, which reduces the effective normal stress and promotes aseismic creep within this region prior to the arrival of the main rupture (Figure 3b). As the rupture approaches this creeping and weakening region, the rupture front experiences a sharp slip rate acceleration, from 0.03 m/s reaching approximately 0.06 m/s (from t1 “black line” to t4 “yellow line” in Figure 3d). This rapid acceleration induces an abrupt stress drop of 5–10 MPa at the rupture front (Figure 3f bottom right zoom-in view, from t1 “black line” to t5 “purple line”) and stress increase behind the rupture front

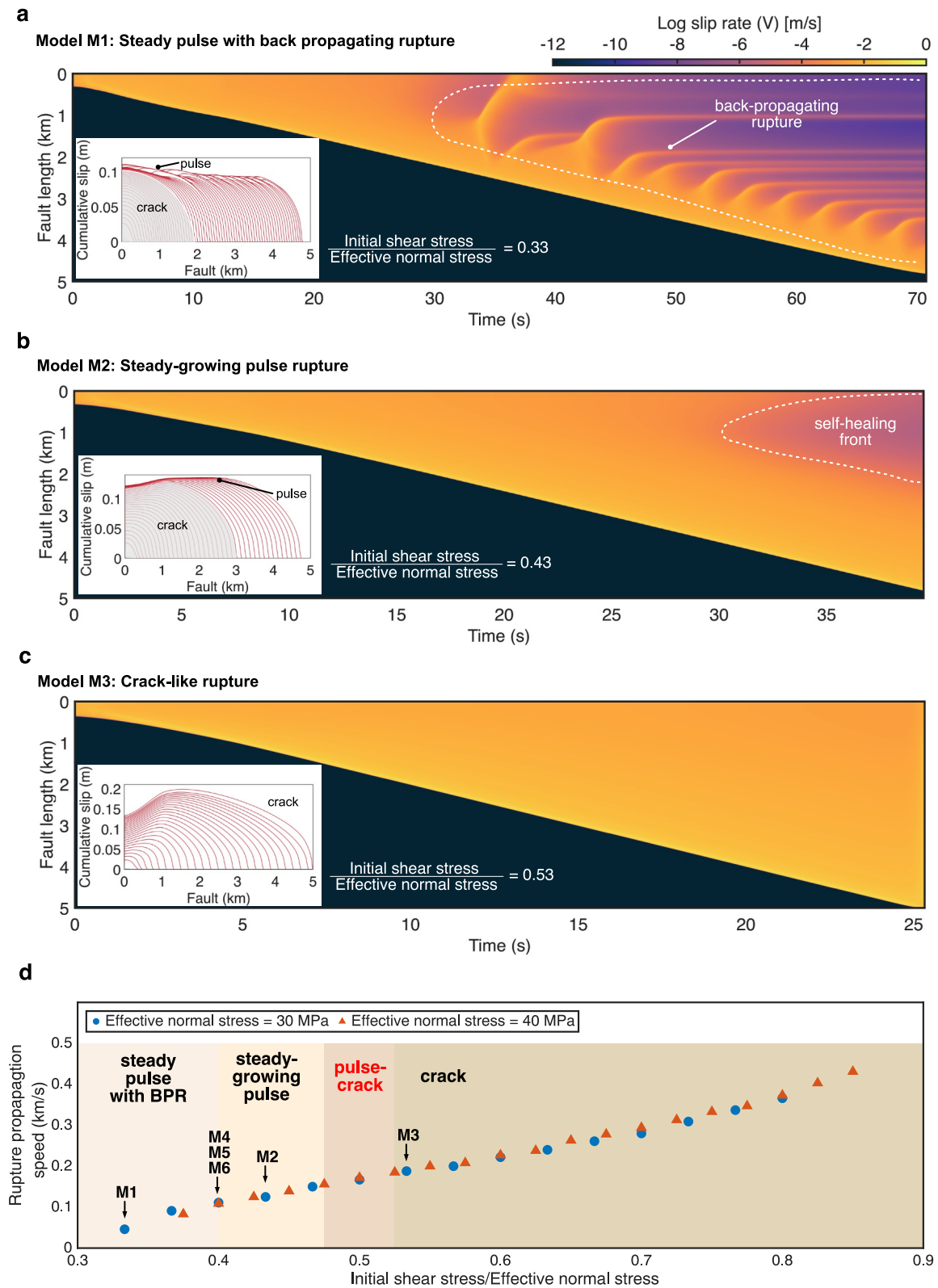


Figure 2.

(Figure 3f bottom left zoom-in view), establishing a pronounced stress gradient between the overpressurized patch and the adjacent, more slowly slipping region. Similar slip acceleration and stress drop have been reported by Sun and Cattania (2025) when the rupture front passes a smooth (low- σ_{eff}) region, although they observed a delayed stress drop, whereas in our simulations the stress drop occurs instantly. The stress gradient drives dynamic unloading and exerts a tensile-like force on the trailing segment, locally increasing slip rate and triggering the onset of BPR. Model M6, incorporating a patch of reduced pore-fluid pressure (i.e., higher effective normal stress), also generates back-propagating rupture (BPR) through a similar and inverse mechanism (Figure S2 in Supporting Information S1).

3.3. Conditions and Mechanisms of BPR

Our simulations indicate that the occurrence of BPR is governed by two primary factors across both homogeneous and heterogeneous faults: a global condition (energy availability) and a local trigger (stress change or interaction with a heterogeneous patch). The global condition determines whether sufficient energy is available for sustained rupture propagation or if re-nucleation occurs, while the local condition governs how stress perturbations or interactions with heterogeneous patches can initiate a secondary rupture.

To evaluate the energy condition, we analyze the evolution of shear stress versus slip on the fault. Figure 4a compares the dynamic and final stress levels of a pulse rupture (Model M1) and a crack rupture (Model M3). The dynamic stress and final stress are comparable for crack rupture. However, in pulse rupture, dynamic stress rapidly increases from 11 MPa due to healing and stress transfer from nearby slipping regions, resulting in a final stress of 13 MPa, which is higher than the dynamic stress. The larger difference between final stress and dynamic stress in pulse rupture than in crack rupture has been confirmed by previous study of Lambert et al. (2021). This elevated residual stress, referred to as “stress undershoot” (Kanamori & Rivera, 2006), maintains additional available energy that can facilitate subsequent re-rupture and trigger BPR.

In earthquake physics, the available energy for rupture propagation, ΔW , is partitioned into radiated energy, E_R , which propagates to the far field, and dissipated energy, with G representing the portion that governs rupture dynamics (Nielsen et al., 2016), also known as fracture energy. Calculation details are provided in the Supporting Information S1. Our results reveal distinct energy partitioning between pulse-like rupture (Model M1) and crack-like rupture (Model M3). Model M3 exhibits substantially larger ΔW and E_R compared to M1, whereas G is comparable (Figure 4b). In this critically stressed scenario, the high energy supply enables rupture to overcome fracture resistance throughout the fault, sustaining continuous crack-like propagation without the need for BPR. In contrast, the lower-stress condition in Model M1 provides insufficient energy to sustain crack growth, leading to rapid slip healing behind the rupture front and leaving residual stress. This condition is favorable for triggering BPR locally.

Even in the presence of residual stress as a necessary condition, the initiation of BPR is controlled by local stress evolution, which determines when and where it nucleates. This triggering process can be interpreted within the framework of Linear Elastic Fracture Mechanics using a simplified analytical model. Details of the calculation are provided in the Supporting Information S1. In this framework, G_c^f and G_c^b are the fracture energy for forward and backward rupture, respectively. They depend on the local shear stress $\tau(x, t)$, which evolves due to frictional healing. As a result, the lower stress level during the healing phase leads to $G_c^b < G_c^f$ (Figure S3 in Supporting Information S1). To relate fracture energy to the stress-based criteria, we express the condition in terms of stress drop. The dynamic stress drops $\Delta\tau^f$ and $\Delta\tau^b$ are measured from the numerical simulations, while the corresponding analytical thresholds $\Delta\tau_{\text{crit}}^f$ and $\Delta\tau_{\text{crit}}^b$ are derived from G_c^f and G_c^b . Accordingly, the condition for forward rupture is $\Delta\tau^f > \Delta\tau_{\text{crit}}^f$, whereas backward rupture requires $\Delta\tau^b > \Delta\tau_{\text{crit}}^b$.

To test whether these analytical criteria capture BPR, we analyze Model M1 (Figures 4c, 4e and 4g) and M5 (Figures 4d, 4f and 4h) as representative examples for homogeneous and heterogeneous fault, respectively. In

Figure 2. Spatio-temporal evolution of rupture and rupture modes. Slip rate (log scale) versus time on the fault for models with different initial shear stress/effective normal stress (a–c), with the bottom-left panel inside showing cumulative slip on the fault. (a) Model M1: Steady pulse rupture with BPR. The dashed-line region marks a self-healing front containing BPRs. (b) Model M2: Steady-growing rupture. (c) Model M3: Crack-like rupture. (d) Parameter study between initial shear stress/effective normal stress and rupture propagation speed. Blue dots represent models with effective normal stress of 30 MPa, and red triangles represent models with effective normal stress of 40 MPa.

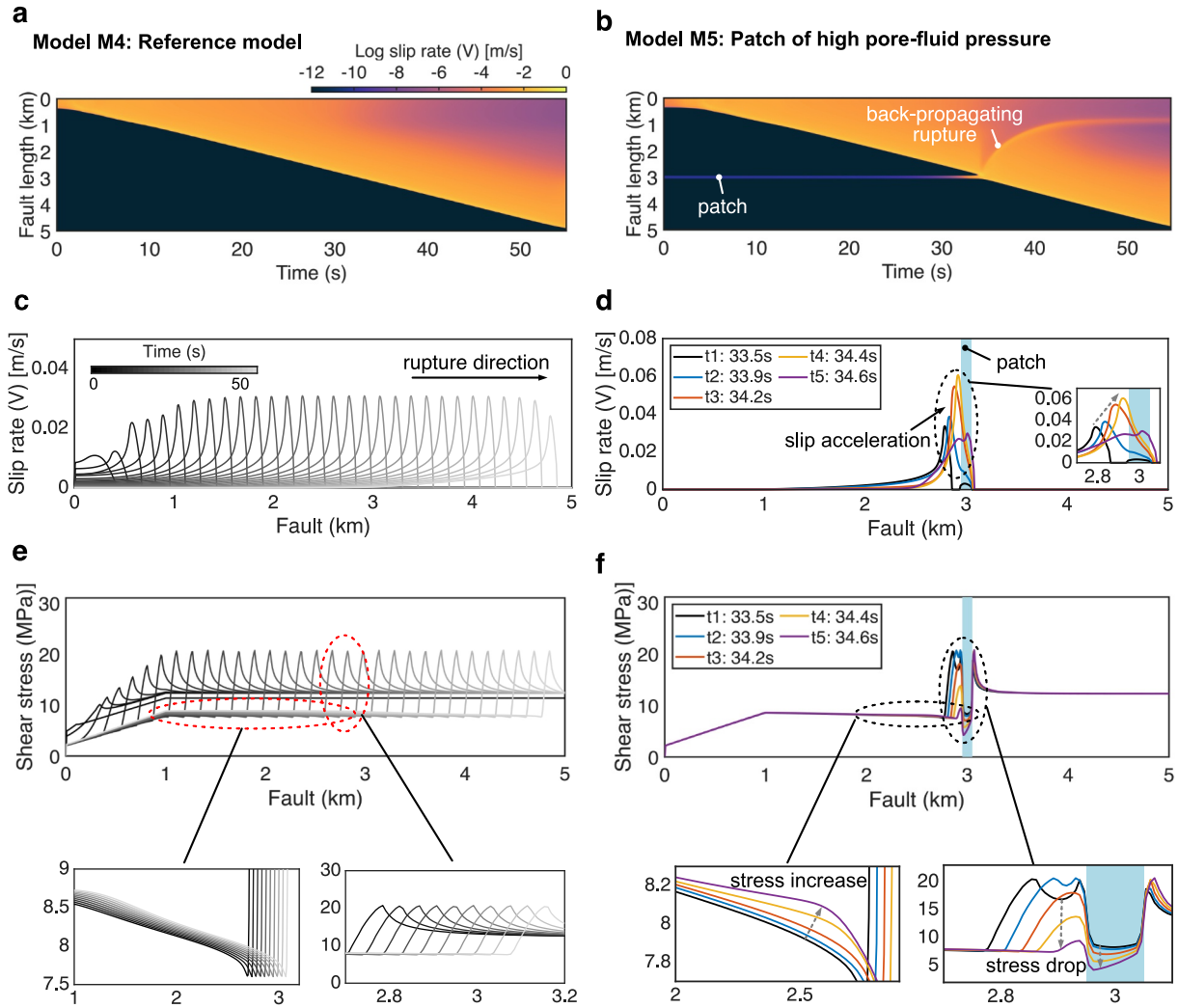


Figure 3. Comparison between homogeneous and heterogeneous models. Left panel: homogeneous fault (reference model). Right panel: model with a patch of high pore-fluid pressure. (a, b) Slip rate (log scale) versus time on the fault. (c, d) Snapshots of slip rate on the fault taken (c) continuously and (d) immediately before BPR. The blue region marks the patch. (e, f) Snapshots of shear stress on the fault. The bottom-left and bottom-right panels show zoomed-in views of shear stress near $x = 3$ km, where BPR initiates in b-f.

Model M1, we identify 11 BPR events (#1—#11), each nucleating at the locations marked by yellow stars and propagating backward (Figure 4c). These BPRs repeatedly nucleate approximately 500 m behind the main rupture front, and each BPR propagates for approximately 400 m until it eventually self-arrests. For example, at fault location 2.8 km, where BPRs #6 nucleates (Figure 4e), shear stress initially increases, then drops, and subsequently increases due to re-strengthening. The forward rupture passes this location at 37 s when the dynamic stress drop $\Delta\tau^f$ reaches the analytical threshold $\Delta\tau_{crit}^f = 7.5$ MPa (Figure 4g). However, the stress drop is incomplete due to the pulse-like rupture driven by a lower τ_0 (as shown in Figure 4a), leaving a residual stress on the fault. Following this phase, the slip rate decreases and the state variable evolves toward larger values (Figure S4 in Supporting Information S1), leading to time-dependent frictional healing and an increase in fault strength. This healing phase enables the fault to be reloaded toward failure, creating favorable conditions for BPR initiation. As a result, slip rate decays below the seismic/aseismic threshold velocity, $V_{th} = 2 a p_{eff} c_s/\mu$ (where c_s is the shear-wave velocity) (Rubin & Ampuero, 2005), marking the onset of BPR at 50.4 s. This threshold velocity serves as a proxy for the initiation of BPR, as rapid frictional restrengthening happens when slip rate falls below V_{th} , consistent with the mechanism proposed by Sun and Cattania (2026), in which BPR initiates at the transition between the fast slipping region near the rupture front and the decelerating region behind. For our parameters, $V_{th} = 1 \times 10^{-3}$ m/s. In our simulations, this critical velocity also coincides with the point at which the energy

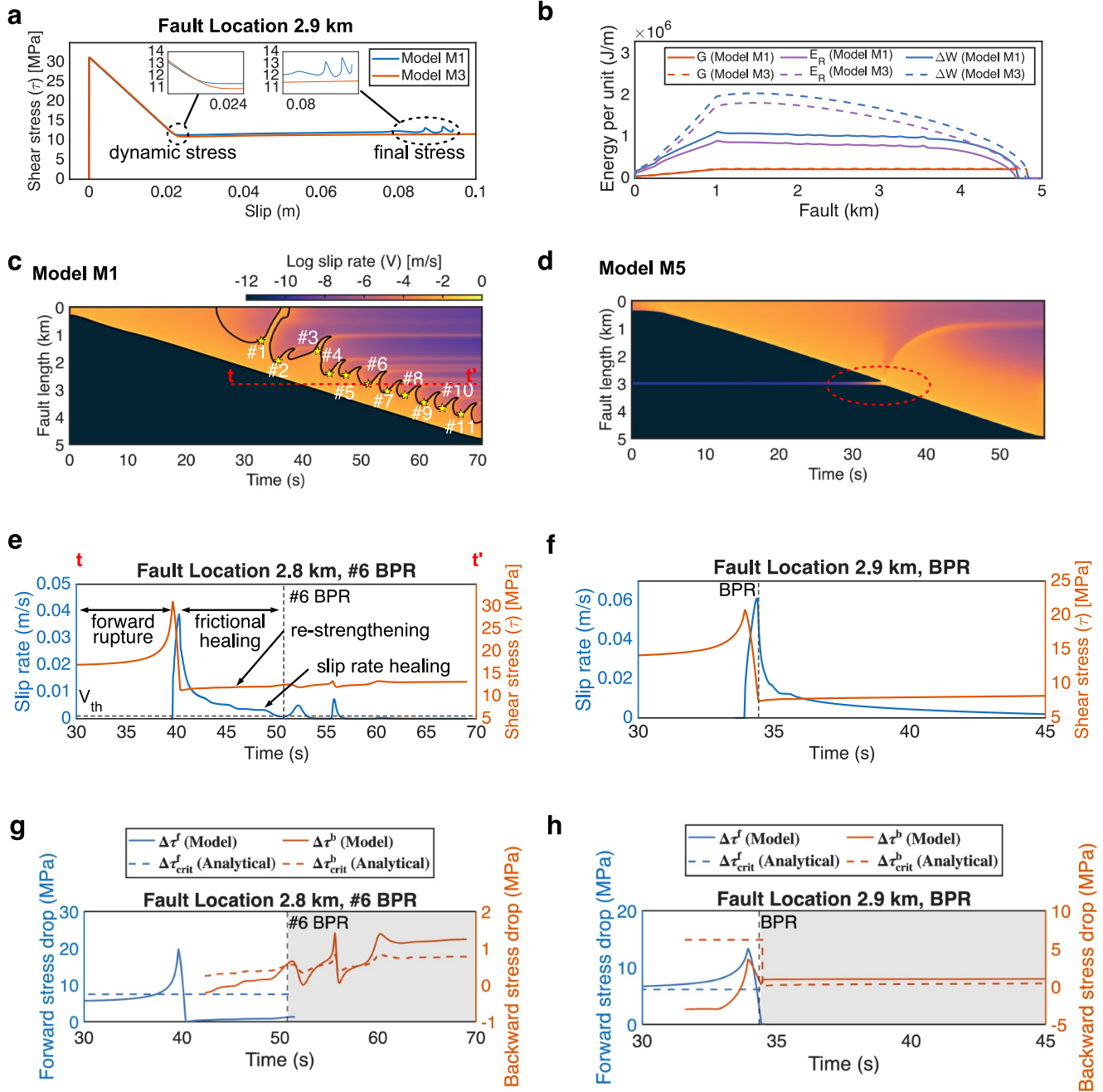


Figure 4. Properties of pulse versus crack rupture in BPR and energy balance analysis. (a) Shear stress versus slip for Model M1 and M3 at 2.9 km, with a zoomed-in view of the dynamic and final stress regions. (b) Energy release on the fault in the form of breakdown energy G , radiated energy E_R , and available energy ΔW for Model M1 and M3. (c, d) Slip rate (log scale) versus time on the fault for Model M1 (c) and Model M5 (d), with black contours representing the region where $V = V_{th}$ and yellow stars marking the start of each BPR. (e) Zoomed-in evolution of slip rate and shear stress for Model M1 at location 2.8 km, corresponding to the nucleation location of #6 BPR. The vertical dashed line marks the onset of BPR, and the horizontal dashed line marks the threshold velocity V_{th} . (f) Evolution of slip rate and shear stress for Model M5 at location 2.9 km. (g, h) Comparison of stress drop between model and analytical calculation for forward (blue lines) and backward (red lines) ruptures in Model M1 (g) and M5 (h).

available for backward rupture is satisfied, suggesting that the threshold V_{th} proposed in Sun and Cattania (2026) reflects the underlying energy balance governing BPR. At this point, the dynamic stress drop $\Delta\tau^b$ reaches the analytical $\Delta\tau_{crit}^b$, which is approximately 0.7 MPa (Figure 4g). This crossing is primarily controlled by the

temporal evolution of shear stress: the forward rupture increases the dynamic stress drop, while subsequent frictional healing reduces the critical threshold for re-rupture.

In Model M5, BPR nucleates when forward rupture encounters the pore-fluid pressure patch at 2.9 km (34.5 s; Figures 4d and 4f). Here, the dynamic stress drop $\Delta\tau^f$ already exceeds $\Delta\tau_{crit}^f = 6$ MPa as the patch is creeping. Upon reaching the patch, the $\Delta\tau_{crit}^b$ immediately drops to 0–0.5 MPa, allowing BPR to initiate almost instantaneously (Figure 4h). The same criterion also explains BPR in the low pore-fluid pressure patch case (Model M6, Figure S2f in Supporting Information S1), demonstrating that the analytical condition is broadly applicable across both homogeneous and heterogeneous faults.

These analytical results demonstrate that the occurrence of BPR is fundamentally controlled by the condition when local stress drop is above the analytical critical threshold for re-rupture on the fault. If the energy condition is met, a subtle stress drop can trigger BPR, whereas forward rupture requires a larger stress drop as shown in Rubin (2011). In homogeneous faults (Model M1), the local stress drop that enables BPR arises from stress recovery (healing) behind the main rupture front. The healing process reduces slip rate below V_{th} and promotes frictional restrengthening, creating stress conditions favorable for the nucleation of BPR. In heterogeneous faults, BPR is induced by transient stress perturbations caused by interaction with a patch of high (Model M5) or low pore-fluid pressure (Model M6).

4. Discussions

Our results reveal a fundamental mechanism for BPR within the framework of energy balance. BPR can be interpreted as a secondary nucleation process facilitated by incomplete stress release, subsequent temporal strength recovery and stress re-distribution. Fault healing plays an important role in this process. In our rate-and-state framework, healing occurs as slip rate decreases following passage of the rupture front, allowing frictional strength to recover. This re-strengthening modifies the local stress and promotes the conditions required for secondary nucleation. The critical thresholds derived above therefore depend on the dynamic stress perturbation. Fluids modulate this process through two mechanisms. First, by controlling the τ_0/σ_{eff} , pore-fluid pressure modifies the initial fault strength and, consequently, the energy available for rupture. Low-energy conditions favor self-healing pulses, leaving regions susceptible to re-nucleation. Second, localized patches of elevated or reduced pore-fluid pressure introduce dynamic heterogeneity, transiently accelerating or decelerating the rupture front and generating stress gradients that can trigger BPR.

To quantify the conditions of BPR, we conduct a parametric study varying both the length of the pore-fluid pressure patch and the ratio of initial shear stress to effective normal stress (τ_0/σ_{eff}). The results indicate that pronounced BPR occurs when τ_0/σ_{eff} is low and the patch is large in the regime of pulse rupture, whereas high τ_0/σ_{eff} and small patches tend to limit BPR in the regime of crack rupture (Figure S5a in Supporting Information S1). This is because, small patches are readily overcome by the rupture with minimal perturbation, while larger patches produce sustained transient stress variations that facilitate BPR. These findings reinforce the global condition that low energy availability favors BPR. This mechanism provides a plausible explanation for secondary rupture in the 2024 Noto earthquake sequence. Since 2020, repeated seismic swarms have ruptured a fluid-rich fault zone (Ma et al., 2024). The mainshock rupture was initially almost arrested by a strong asperity (with high stress), but this asperity was subsequently broken after a second rupture nucleated on the opposite side and propagated backward across the asperity (Xu et al., 2024). This sequence closely matches the behavior observed in our Model M5, in which a strong heterogeneity interacts with rupture and is later overcome by the triggered BPR. Similar phenomenon has also been observed in Episodic Tremor and Slip in Cascadia (Ghosh et al., 2010), where the fluid heterogeneity perturbs the effective normal stress on the fault, which triggers shear failure that propagates rapidly in the backward direction and produces the observed back-propagating tremor.

The observed BPRs are typically classified as subshear to supershear ruptures (Ding et al., 2023; Hicks et al., 2020; Hu et al., 2021; Lee et al., 2011; Okuwaki et al., 2023; Uchide et al., 2013; Wei et al., 2011; Xu et al., 2024; Yao et al., 2020) or RTRs (Ghosh et al., 2010; Houston et al., 2011; Nakamoto et al., 2021; Obara et al., 2012; Plourde et al., 2015; Royer et al., 2015; Thomas et al., 2013; Yamashita et al., 2015), propagating on the order of 1 km/s and 0.001 km/s (100 km/day), with stress drop of 1–10 MPa and 0.1–10 kPa, respectively (Figure S5b in Supporting Information S1). The lower rupture velocity and stress drop from the model compared to natural large earthquakes likely results from the relatively low background stress used in the model. In addition,

we do not observe supershear BPR in our simulations, despite previous numerical studies demonstrating that both subshear and supershear back-propagating ruptures can arise from a variety of mechanisms (Ding et al., 2024; Flores-Cuba et al., 2024; Sun & Cattania, 2026). These limitations may result from the assumption of a single, simplified fault segment, which neglects interactions among multiple segments and structural features such as fault bends (Abdelmeguid et al., 2023), fault branches (Ding et al., 2023; Fliss et al., 2005; Jia et al., 2023; Oglesby et al., 2003; Xu et al., 2015), step-overs (Mia et al., 2024), and sub-parallel faults (Li et al., 2016; Tanaka et al., 2014), all of which have been proposed as contributors to BPR in large earthquakes.

Although our model does not capture the full rupture-speed spectrum, the underlying mechanics remain relevant, as rupture initiation and propagation in our models are governed by energy balance. Therefore, the physical triggering mechanism of BPR is expected to hold across timescales and contexts. We acknowledge that poro-elastic effects (e.g., permeability evolution and pore-fluid pressure diffusion) are not explicitly resolved in this study, as their characteristic timescales (days to years) are much longer than those of dynamic rupture (seconds) (Miller, 2002). Instead, pore-fluid pressure primarily acts to set the effective normal stress, thereby controlling the $\tau_0/\sigma_{\text{eff}}$ conditions and introducing spatial heterogeneity along the fault. Future work should investigate the combined roles of fluids and BPR in more realistic fault networks, where fluid migration interacts with complex geometry. Extending simulations to 3D would further capture dynamic processes within fault zones. Moreover, high-resolution earthquake catalogs and back-projection analyses could help identify natural analogs of BPR by resolving spatiotemporal patterns of secondary nucleation and reverse rupture migration. Specifically, dense catalogs can detect closely spaced events that migrate opposite to the main rupture direction, while back-projection can image backward-propagating high-frequency energy fronts. Together, these observations can constrain how stress state and fluid-driven stress perturbations control the occurrence of BPR in complex rupture sequences.

5. Conclusions

This study highlights the important role of fluids in shaping the onset and evolution of BPR. We demonstrate that low $\tau_0/\sigma_{\text{eff}}$ conditions favor pulse-like rupture, in which incomplete stress release and temporal healing facilitate secondary nucleation behind the rupture front. Localized pore-fluid pressure patches further modulate slip rates and stress distributions, promoting BPR when local stress drop exceeds the analytical critical threshold for re-rupture. By integrating numerical modeling with insights from natural observations, we identify how energy availability and fluid-induced heterogeneities contribute to complex rupture patterns. Despite the simplified model setup, this work provides a framework for understanding the interplay between fluids and complex fault rupture dynamics.

Conflict of Interest

The authors declare no conflicts of interest relevant to this study.

Availability Statement

All simulations were performed in the open-source code H-MECs (Dal Zilio & Computational, 2026). The simulation data in this study are available at Ye, Dal Zilio, and Giardini (2026).

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